

Archived v2 documentation. Use the explicit `--historical-v2` launch flag described in [current instructions](#); the default launch now uses candidate data.

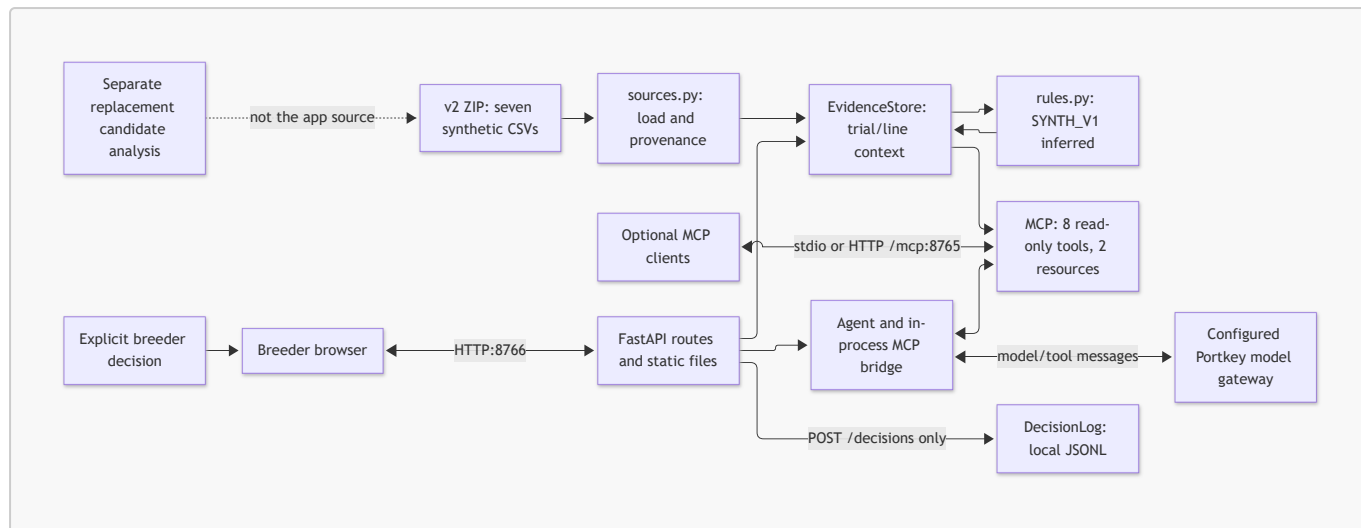
UC4 implemented architecture

Current implementation reviewed 2026-09-30. The demo uses the **v2** synthetic archive. The [current replacement candidate analysis](#) uses different schemas and candidate-level GREEN/AMBER/RED; it is not implemented by this historical v2 app. The [v3 analysis](#) is superseded historical evidence. See [review evidence and readiness](#), [v2 data model](#), and [run instructions](#).

Components and stack

Component	Implementation	Responsibility
Runtime/package management	Python >=3.12, uv; declared ranges in pyproject.toml	Local console entry points <code>uc4-mcp</code> and <code>uc4-ask</code> ; resolved versions recorded per review run.
Source and evidence	zipfile, pandas; sources.py and store.py	Read seven CSVs directly from a ZIP; retain CSV member, row ID and line number; construct trial/line context and consistency flags.
Scoring	rules.py, SYNTH_V1	Infer fixed cut-points; deterministic PASS/HOLD/FAIL at trial scope, with five PASS criteria and two knockouts.
MCP	mcp SDK; server.py	Eight read-only tools and two JSON resources. stdio or streamable HTTP <code>/mcp</code> on 8765.
Agent	bridge.py, agent.py, grounding.py, llm.py	In-process MCP client; OpenAI-compatible client through configured Portkey gateway; bounded tool rounds, citations and repair.
Web/API	FastAPI, uvicorn; api.py	Static breeder page, trial/line reads, Ask, explicit decisions; default localhost:8766.
Browser	HTML, CSS, JavaScript	Render evidence, clarify identifiers, show answers and submit explicit breeder choices.
Decisions	decisions.py, models.py	Local JSONL append with copied recommendation, reason, demo alias and UTC timestamp; separate from source data.

No database, training pipeline, n8n workflow, cloud deployment, or live research-system integration is implemented here. OpenCode, Claude Code and n8n are documented optional MCP clients.



Data, scoring and provenance

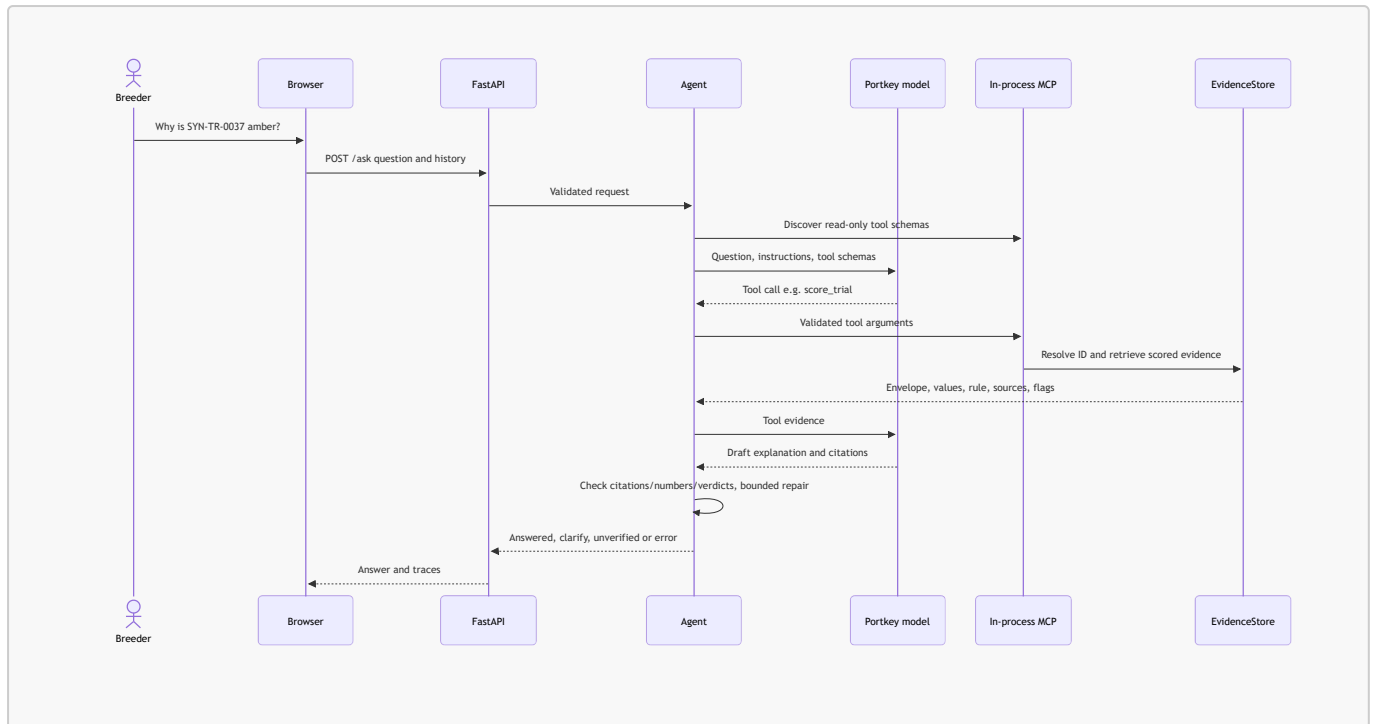
The exact v2 default filename is `get_started/RE__Hatchworks_Hackathon_-_4th_Use_Case.zip`. `UC4_ZIP` explicitly overrides it; it does not migrate the schema. Version compatibility is checked by the loader. Archives are git-ignored and must be obtained from the supplied local materials or a teammate with approved access.

Recommendations join trials by `TRIAL_GUID`. v2 observations link trials by `ATTACHED_TO_FIELD_ENTITY_ID` and lines by `GID`; they contain no field trait measurements. Genomics/lab attach to lines by `MATERIAL_GUID`; lab has no trial key or trait dictionary. Trial aggregate values come from recommendations, not a reconstruction of observation-linked lines. Genomics reconciliation flags expose discrepancies rather than repair the supplied membership silently.

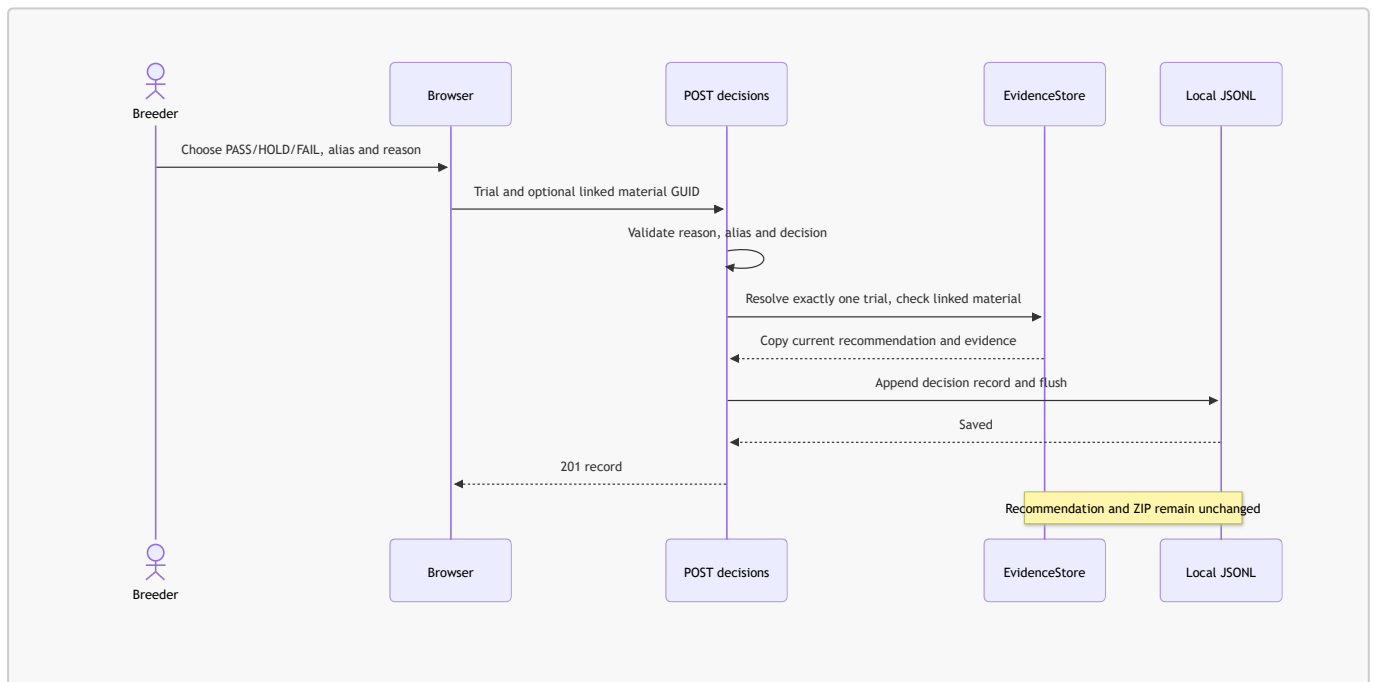
The inferred rule is FAIL for yield < 7 t/ha or disease > 7; otherwise PASS requires yield >= 9, moisture <= 22%, disease <= 5, mean GBV >= 102, and resistant material >= 50%; otherwise HOLD. Missing criteria cannot earn PASS. Height and flowering are shown as context and do not set this score. Matching the synthetic baseline does not confirm thresholds or validate a biological advancement decision.

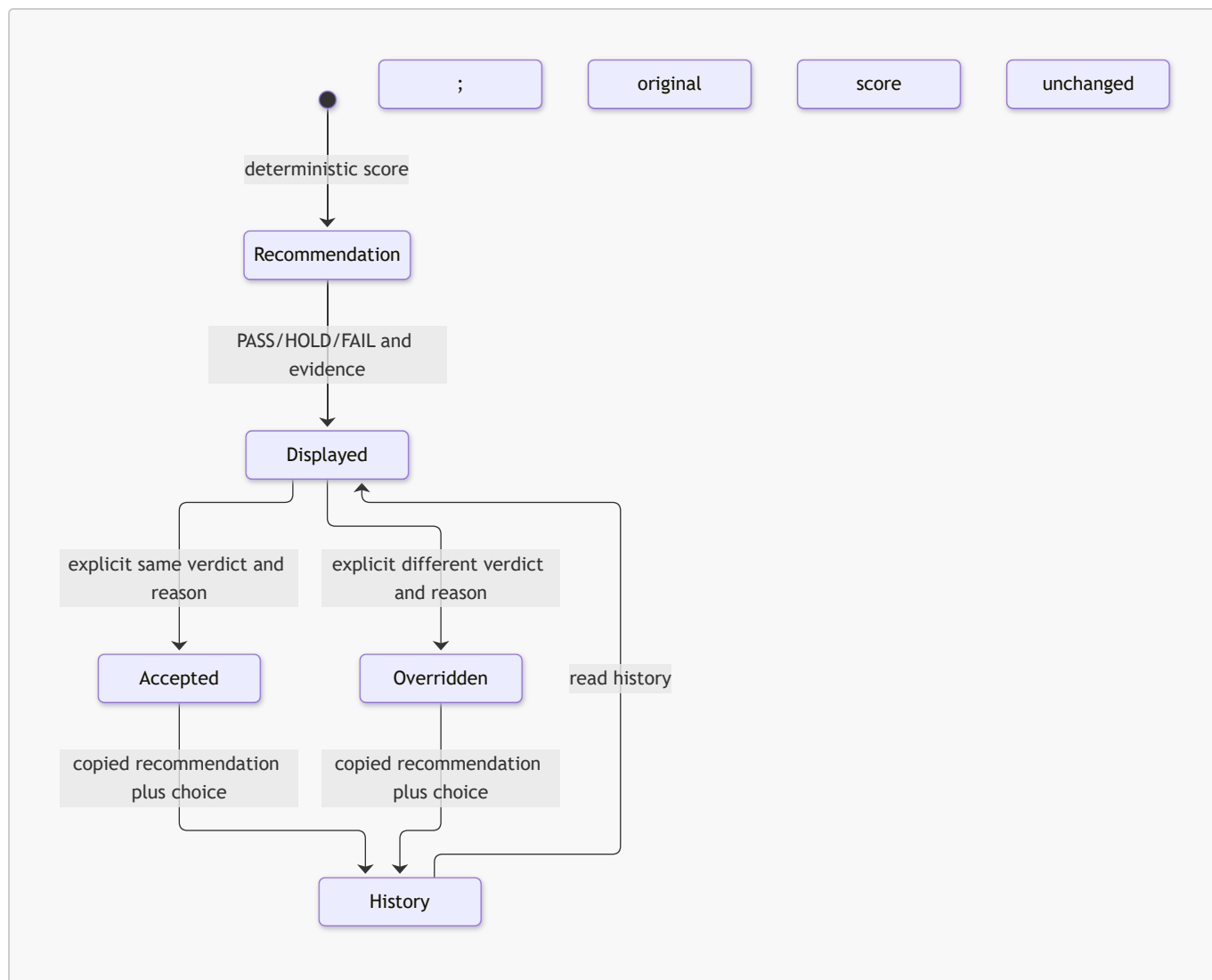
CSV evidence is cited as `[source_file#row_id]`; aggregate tool results can use `[tool:query_trials]` or another tool reference. Derived criteria carry rule/threshold information. These are different sources of justification and must remain identifiable.

Question and decision flows



Ambiguous identifiers return candidate lists; absence stays missing. Numeric/citation checks provide a guard, not a full semantic proof: a real value may still be attributed to the wrong trait/entity, and a supported answer may omit part of the request. Review evidence includes semantic checks. Only retrieved evidence supports answer quantities; the question itself is not a factual numeric source.





Decision records contain `decision_id`, `trial_guid`, `trial_id`, optional `material_guid`, the copied `recommendation`, `decision`, `overrides`, `reason`, `user`, and server UTC `timestamp`. Optional material only annotates a trial decision; it is not a per-line score. `UC4_DECISION_LOG` overrides the default `app/data/decisions.jsonl`. Rehearsals use fresh isolated logs.

Assumptions, limits and records

The server has no authentication. `user` is a demo alias, and JSONL is append-only through this API; it is neither tamper-proof nor a multi-process transactional audit database. No model/tool path records a breeder decision. Binding to all interfaces is a separately documented local-client choice, not a production deployment.

Business/source assumptions and technical choices are documented in [review assumptions](#) and [technical decisions](#). Each technical choice records context, alternatives, evidence, decision-maker, consequences, and supersession. Breeder decisions are instead the application JSONL records described above. [SME answers](#) record external statements and unresolved questions; [prompt log](#) records retained AI development prompts.

Review and reproduction

Default pytest excludes browser/live checks. Run analysis, offline application, trusted browser input and live-model evaluation as separate lanes, preserving failures/skips. Follow [demo review tasks](#) for source-to-screen

comparisons, video provenance, and final readiness. Historical [Phase 3 evidence](#) remains dated historical evidence. Recordings supplement the handbook-required live event presentation.